BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 4 | Duration: 40 min
Semester: Spring 2025 | Full Marks: 15

CSE 470: Software Engineering

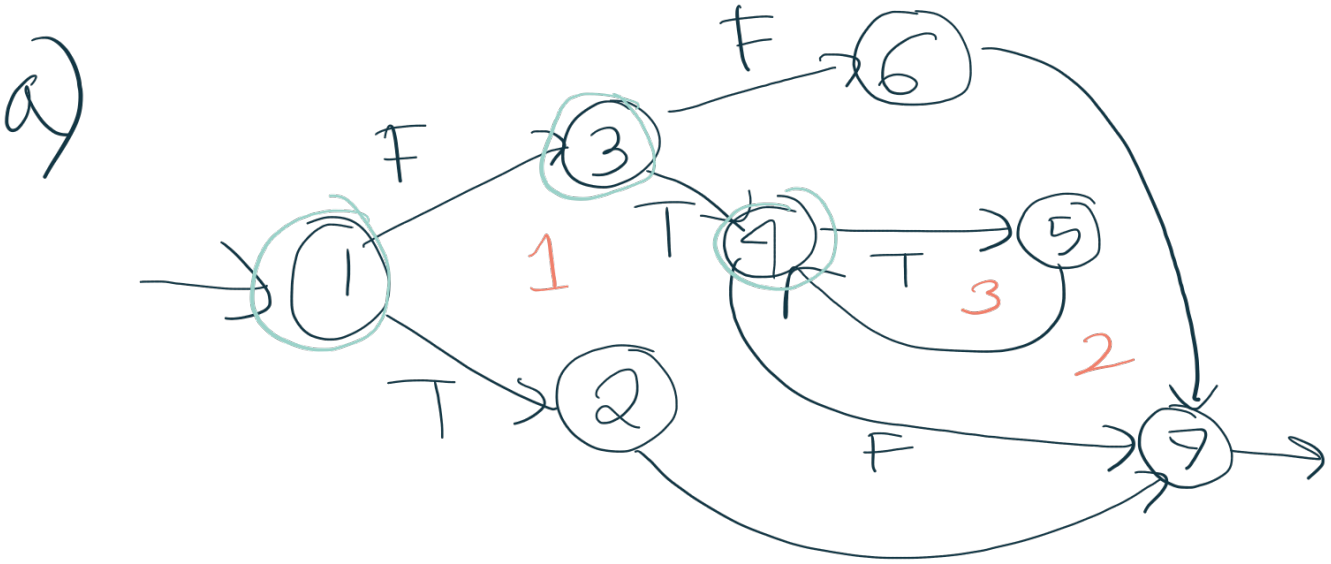
Name: **ID:** **Section:**

Python	Java

user = "admin"
passw = "1234"
tries = 0
 	
if **i_user** != user:
  print("User not found.")
elif **i_pass** != passw:
  while tries < **limit**:
  print(f"Wrong password. Attempt {tries + 1}")
  tries += 1
else:
  print("Access granted.")
	public class Login {
	  public void login(String iUser, String iPass, int limit){
	  String user = "admin";
	  String passw = "1234";
	  int tries = 0;
	
	  if (!iUser.equals(user)) {
	  System.out.println("User not found.");
	  }
	  else if (!iPass.equals(passw)) {
	  while (tries < limit) {
	  System.out.println("Wrong password. Attempt " + (tries + 1));
	  tries++;
	  }
	  }
	  else {
	  System.out.println("Access granted.");
	  }
	  }
	}

[You must mark which code block/line corresponds to which node in the question paper.]

1.	**CO3**	a. Draw the CFG for the given method [5 Marks] b. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set [3 + 3 = 6 Marks] c. Generate test cases for all paths in the Basic Paths Set [4 Marks]	**15**



b) $M = R + 1 = 3 + 1 = 4$

$P + 1 = 3 + 1 = 4$

$E - N + 2P = 9 - 7 + 2(1)$

$= 4$

Paths Set :

① 1 – 2 – 7

② 1 – 3 – 4 – 5 – 4 – 7

③ 1 – 3 – 6 – 7

c) Test case:

① iUser = "wrong"

Output : "User not found"

② iUser = "admin" iPass = "wrong"

limit = 1

Output : Wrong password. Attempt 1

③ iUser = "admin" iPass = "1234"